

DD_SUB-WF-TERMINATE-01 - Subscription Termination

Developer implementation guide: DD_SUB-WF-TERMINATE-01-DEV_IMPL_GUIDE-v1.0.md.

1. Purpose

SUB-WF-TERMINATE-01 defines permanent subscription termination.

Termination moves a subscription to:

TERMINATED

TERMINATED is a terminal state. A terminated subscription must not be activated, paused, resumed, suspended, or restricted again. A new service for the customer must use a new subscription.

Use this operation when:

- A customer requests cancellation.
- A BO admin closes a subscription.
- BIL-04 reaches final dunning termination.
- A contract-expiry scheduler terminates an expired contract.

2. Mental Model

Termination is a saga:

```
flowchart LR
    API[Terminate endpoint] --> INT[Interrupt in-flight operation]
    INT --> FW[SUB-WF framework]
    FW --> BPMN[Terminate BPMN]
    BPMN --> RULES[Drools validation]
    BPMN --> PEND[SUB-LM PENDING_TERMINATION]
    BPMN --> BIL[Billing termination intent]
    BPMN --> FUL[FUL-05 revoke service]
    BPMN --> HP[Release HomePass]
    BPMN --> LM[SUB-LM TERMINATED]
    BPMN --> EVT[Kafka event]
    EVT --> RMA[Equipment collection]
    BPMN --> NOTIF[Notification]
```

The most important output is not only the status. It is the combination of:

- final status TERMINATED
- termination master fields on subscription
- billing settlement
- network/service revocation
- HomePass release
- equipment disposition
- SubscriptionTerminated event

3. Ownership

This DD owns:

- Termination trigger API: POST /api/subscriptions/{subscription_id}/terminate.
- Termination preview API: POST /api/subscriptions/{subscription_id}/terminate/preview.
- Termination trigger taxonomy.
- Equipment disposition taxonomy.
- subscription_termination_config table.
- Termination business rules.
- Termination domain events.

- Termination worker usage.

This DD does not own:

- WIK-specific BPMN. See DD_SUB-WF-TERMINATE-01-FLOW-WIK.
- Subscription master state implementation. See SUB-LM-01.
- Operation orchestration framework. See SUB-WF-FRAMEWORK-01.
- Worker topic contracts. See DD_SUB-WF-FRAMEWORK-01-WORKERS.
- Billing settlement calculation. See BIL-01.
- Dunning final-decision logic. See BIL-04.
- Network revocation implementation. See FUL-05.
- HomePass release implementation. See FUL-07.
- Equipment pickup work orders. See OSR-RMA-01 EQP.
- Notification templates and delivery. See NOT-01.

4. Deployment Unit Clarification

Termination is not a standalone service.

It is deployed as part of the subscription workflow capability:

- Terminate route/controller in SUB-WF API.
- Termination config migration.
- Termination BPMN, such as sub-terminate-wik.
- Termination Drools package, such as rules.subscription.terminate.wik.
- Shared worker apps from the worker catalog.

The same worker app can execute termination, resume, pause, and activation topics. Keep metrics, logs, retries, and alerts visible by topic.

5. Termination Triggers

Trigger	Meaning	Allowed roles
CUSTOMER_REQUESTED	Customer requests cancellation through portal, call center, or operator.	CUSTOMER_SELF_SERVICE, CALL_CENTER, SUBSCRIPTION_OPERATOR
ADMIN_DECISION	Admin/operator terminates for operational, compliance, or exceptional reason.	SUBSCRIPTION_ADMIN
DUNNING_TERMINATION	Billing terminates after final non-payment dunning step.	BILLING_INTERNAL
CONTRACT_EXPIRED	Scheduler terminates an expired fixed-term contract.	SCHEDULER_INTERNAL

CUSTOMER_SELF_SERVICE also requires:

- `subscription_termination_config.customer_self_service_enabled = true`

6. Equipment Disposition

Equipment disposition tells downstream systems what to do with customer-premise equipment.

Disposition	Meaning	Typical downstream behavior
CUSTOMER_RETAINS	Customer keeps equipment.	No collection work order.
PENDING_COLLECTION	Equipment must be collected.	OSR-RMA-01 creates pickup work order.
WRITTEN_OFF	Equipment is written off.	No collection attempt, finance/reporting audit.

Resolution rules:

- Caller may supply equipmentDisposition.
- If absent and trigger is not DUNNING_TERMINATION, use default_equipment_disposition.
- If absent and trigger is DUNNING_TERMINATION, use dunning_equipment_disposition.
- If dunning auto write-off threshold is configured, BIL-01 may return a writeOffMarker; BPMN may resolve disposition to WRITTEN_OFF.

7. Data Model

7.1 subscription_termination_config

What this table is for:

subscription_termination_config stores operator-level termination policy. It controls cooling-off behavior, default equipment handling, dunning write-off behavior, self-service enablement, supervisor override, and churn-save routing.

Use this table when:

- Validating customer self-service termination.
- Resolving default equipment disposition.
- Deciding whether dunning termination can auto write off debt/equipment.
- Computing cooling-off window.
- Deciding whether supervisor override or churn-save step is needed.

Do not use this table for:

- Current subscription status.
- Actual billing settlement calculation.
- Actual equipment inventory ownership.
- Network revocation decisions.

```
CREATE TABLE subscription_termination_config (
  operator_code          TEXT PRIMARY KEY,
  cooling_off_window_days INTEGER NOT NULL DEFAULT 7,
  default_equipment_disposition TEXT NOT NULL DEFAULT 'PENDING_COLLECTION',
  dunning_equipment_disposition TEXT NOT NULL DEFAULT 'PENDING_COLLECTION',
  dunning_auto_write_off_threshold NUMERIC(18,2) NULL,
  dunning_auto_write_off_currency TEXT NULL,
  customer_self_service_enabled BOOLEAN NOT NULL DEFAULT FALSE,
  cooling_off_supervisor_override BOOLEAN NOT NULL DEFAULT FALSE,
  churn_save_attempt_enabled BOOLEAN NOT NULL DEFAULT FALSE,
  updated_at             TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  updated_by             TEXT NOT NULL,
  CHECK (default_equipment_disposition IN ('CUSTOMER_RETAINS', 'PENDING_COLLECTION', 'WRITTEN_OFF')),
  CHECK (dunning_equipment_disposition IN ('CUSTOMER_RETAINS', 'PENDING_COLLECTION', 'WRITTEN_OFF')),
  CHECK ((dunning_auto_write_off_threshold IS NULL) = (dunning_auto_write_off_currency IS NULL))
);
```

Column descriptions:

Column	Purpose
operator_code	Operator this config applies to.
cooling_off_window_days	Number of days after activation where customer cancellation may receive cooling-off treatment.
default_equipment_disposition	Equipment handling when caller does not supply one for non-dunning termination.
dunning_equipment_disposition	Equipment handling when caller does not supply one for dunning termination.
dunning_auto_write_off_threshold	Debt threshold used to decide whether dunning termination can mark write-off.
dunning_auto_write_off_currency	Currency for the dunning write-off threshold.
customer_self_service_enabled	Allows customer portal/app to start customer-requested termination.
cooling_off_supervisor_override	Adds supervisor decision step for cooling-off cases.
churn_save_attempt_enabled	Adds a churn-save attempt before customer-requested termination proceeds.
updated_at	Last update timestamp.

Column	Purpose
updated_by	User/service that last updated the row.

Sample data:

operator_code	cooling_off_win dow_days	default_equipm ent_disposition	dunning_equip ment_dispositio n	dunning_auto_ write_off_thres hold	dunning_auto_ write_off_curre ncy	customer_self_ service_enable d	cooling_off_su pervisor_overri de	churn_save_att empt_enabled
WIK	7	PENDING_COL LECTION	PENDING_COL LECTION	10000.00	KES	true	true	false
WUG	7	PENDING_COL LECTION	PENDING_COL LECTION	NULL	NULL	false	false	false
WTZ	7	PENDING_COL LECTION	PENDING_COL LECTION	NULL	NULL	false	false	false
YASSN	7	PENDING_COL LECTION	PENDING_COL LECTION	NULL	NULL	false	false	false

Sample meaning:

- WIK allows self-service termination.
- WIK applies a 7-day cooling-off policy.
- WIK default equipment handling is collection.
- WIK may auto write off dunning debt/equipment up to KES 10,000 according to billing policy.
- WIK does not add a churn-save BPMN step in v1.0.
- Other operator rows are conservative placeholders until their flow satellites are defined.

7.2 Subscription Termination Master Fields

What this data is for:

Termination writes final audit fields onto the subscription master record. SUB-LM owns the actual subscription table and state machine.

Use this data when:

- Showing support why a subscription was terminated.
- Preventing future lifecycle operations on a terminated subscription.
- Building events and reports.
- Auditing dunning/admin/customer termination.

Fields written by termination:

Field	Purpose	Sample
terminated_at	Actual termination timestamp.	2026-07-20T14:30:18Z
termination_trigger	Trigger that started termination.	CUSTOMER_REQUESTED
termination_reason_code	Operator/CVM reason.	CUSTOMER_MOVED_NO_COVERAGE
prior_status_at_termination	Status before termination saga started.	ACTIVE
contract_started_at	Contract start snapshot.	2024-01-20T10:00:00Z
contract_end_at	Contract end snapshot, if applicable.	2026-01-20T10:00:00Z
contract_minimum_term_months	Contract minimum term.	12

8. Business Rules

8.1 State And Concurrency Rules

Rule	Description
R-TERM-S-1	Termination rejects subscriptions already TERMINATED with 409 ALREADY_TERMINATED.

Rule	Description
R-TERM-S-2	Termination is terminal. No future lifecycle operation may mutate the subscription after success.
R-TERM-S-3	Termination may interrupt another in-flight state operation on the same subscription.
R-TERM-S-4	If another operation is in-flight, framework sends <code>external-cancel-requested</code> , waits up to 60 seconds, then starts termination after cancellation acknowledgement.
R-TERM-S-5	During the saga, subscription moves to <code>PENDING_TERMINATION</code> before final <code>TERMINATED</code> .

8.2 Trigger Rules

Rule	Description
R-TERM-T-1	Trigger must be one of the four values in section 5.
R-TERM-T-2	Actor role must match the trigger role set.
R-TERM-T-3	<code>CUSTOMER_SELF_SERVICE</code> requires operator config enabled.
R-TERM-T-4	<code>terminationReasonCode</code> is required and passed through to reporting/CVM.

8.3 Cooling-Off Rules

Rule	Description
R-TERM-C-1	Cooling-off applies only to <code>CUSTOMER_REQUESTED</code> .
R-TERM-C-2	A termination is in cooling-off when <code>activated_at + cooling_off_window_days >= requestTime</code> .
R-TERM-C-3	Cooling-off result is sent to BIL-01 in <code>TERMINATION_INTENT</code> .
R-TERM-C-4	If <code>cooling_off_supervisor_override = true</code> , BPMN includes a supervisor decision step.

8.4 Settlement Rules

Rule	Description
R-TERM-B-1	BPMN sends <code>TERMINATION_INTENT</code> to BIL-01.
R-TERM-B-2	BIL-01 returns line items, totals, refund amount, early termination fee, and write-off marker.
R-TERM-B-3	Settlement result is included in <code>SubscriptionTerminated</code> .
R-TERM-B-4	Termination does not hand-code billing amounts in SUB-WF.

8.5 Fulfillment And Equipment Rules

Rule	Description
R-TERM-F-1	BPMN calls FUL-05 to revoke network/service access.
R-TERM-F-2	BPMN calls HomePass release worker after service revoke succeeds.
R-TERM-F-3	Equipment bindings are passed to FUL-05 and included in the event.
R-TERM-F-4	OSR-RMA-01 creates pickup work order from <code>SubscriptionTerminated</code> when disposition is <code>PENDING_COLLECTION</code> .

9. REST APIs

9.1 Trigger Termination

```
POST /api/subscriptions/{subscription_id}/terminate HTTP/1.1
Authorization: Bearer <jwt-or-service-token>
Idempotency-Key: term-sub_01HZ_KAMAU-20260720T143000-cc007
Content-Type: application/json
```

```
{
  "operatorCode": "WIK",
  "terminationTrigger": "CUSTOMER_REQUESTED",
```

```

    "terminationReasonCode": "CUSTOMER_MOVED_NO_COVERAGE",
    "equipmentDisposition": "PENDING_COLLECTION",
    "customerNotes": "Moving to a non-covered area; no service available at new address",
    "initiatingWorkflowKind": "BO_UI",
    "initiatingWorkflowRef": "cc-ticket-2026-07-20-4421"
  }
}

```

Actor user and role come from the token, not the public request body.

Success:

202 Accepted

```

{
  "operationId": "subop_01HZ_KAMAU_TERM",
  "operatorCode": "WIK",
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "operationKind": "TERMINATE",
  "terminationTrigger": "CUSTOMER_REQUESTED",
  "bpmnProcessKey": "sub-terminate-wik",
  "currentState": "INITIATED",
  "finalState": null
}

```

Common errors:

HTTP	Code	Meaning
400	UNKNOWN_TERMINATION_TRIGGER	Trigger is not recognized.
400	UNKNOWN_EQUIPMENT_DISPOSITION	Equipment disposition is not recognized.
400	TERMINATION_REASON_REQUIRED	Reason code missing.
403	ROLE_NOT_AUTHORIZED_FOR_TRIGGER	Actor role cannot use this trigger.
403	CUSTOMER_SELF_SERVICE_NOT_ENABLED	Customer self-service termination disabled.
409	ALREADY_TERMINATED	Subscription is already terminal.
409	OPERATION_INTERRUPTION_FAILED	Existing in-flight operation did not cancel in time.
409	IDEMPOTENCY_KEY_REUSED	Same idempotency key was reused with different request hash.

9.2 Termination Preview

Preview is read-only. It validates request shape and asks dependent modules for estimated settlement and equipment context. It does not start BPMN and does not change status.

```

POST /api/subscriptions/{subscription_id}/terminate/preview HTTP/1.1
Authorization: Bearer <jwt-or-service-token>
Content-Type: application/json

```

```

{
  "operatorCode": "WIK",
  "terminationTrigger": "CUSTOMER_REQUESTED",
  "terminationReasonCode": "CUSTOMER_MOVED_NO_COVERAGE",
  "equipmentDisposition": "PENDING_COLLECTION"
}

```

Response:

```

{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "coolingOffApplicable": false,
  "daysActiveInContract": 920,
  "contractMinimumTermMonths": 12,
  "equipmentDisposition": "PENDING_COLLECTION",
  "equipmentBindings": [
    {
      "equipmentRequirement": "GPON_ONT",
      "equipmentRef": "ont_01HZ_KAMAU"
    },
    {
      "equipmentRequirement": "ATA",
      "equipmentRef": "ata_01HZ_KAMAU"
    }
  ],
  "settlement": {
    "lineItems": [
      {
        "kind": "PRORATED_REFUND",
        "amount": -1200.00,

```

```

        "currency": "KES",
        "description": "Refund for 11 unused days of July cycle"
    }
},
"totals": {
    "net": -1200.00,
    "currency": "KES"
},
"refundAmount": 1200.00,
"writeOffMarker": false
}
}

```

10. Worker Usage In This Operation

The worker list is a logical BPMN task list, not a deployment list.

Worker topic	Why termination uses it
drools-decide	Validate state, trigger, role, cooling-off, disposition, and routing flags.
sub-lm-cancel-in-flight-operation	Interrupt an existing in-flight operation when termination takes precedence.
sub-lm-put-pending-status	Move to PENDING_TERMINATION.
sub-lm-compute-cycle-window	Build billing context.
bil01-emit-intent	Ask BIL-01 to compute TERMINATION_INTENT settlement.
ful-call-terminate	Call FUL-05 to revoke service.
ful-call-release-homepass	Release HomePass slot through FUL-07 or approved stub.
sub-lm-put-master-fields	Write termination master fields and audit data.
sub-lm-commit-state	Commit subscription to TERMINATED.
sub-lm-revert-to-prior	Compensation before terminal commit, if safe.
kafka-emit	Emit termination success/rejected event.
notification-send	Notify customer/admin, reliability-isolated.
subscription-operation-finalize	Mark operation terminal.

11. Kafka Events

11.1 SubscriptionTerminated

Topic:

```
sophix.subscription.subscription.terminated
```

Emitted after:

- FUL-05 revocation succeeds.
- HomePass release succeeds or approved stub succeeds.
- SUB-LM commits status to TERMINATED.

Sample:

```

{
  "eventType": "SubscriptionTerminated",
  "aggregateId": "sub_01HZ_KAMAU_FIBER",
  "timestamp": "2026-07-20T14:30:18Z",
  "payload": {
    "operatorCode": "WIK",
    "subscriptionId": "sub_01HZ_KAMAU_FIBER",
    "customerId": "cust_01HZ_KAMAU",
    "homepassId": "hp_01HZ_KAREN_017",
    "operationId": "subop_01HZ_KAMAU_TERM",
    "terminationTrigger": "CUSTOMER_REQUESTED",
    "terminationReasonCode": "CUSTOMER_MOVED_NO_COVERAGE",
    "priorStatus": "ACTIVE",
    "newStatus": "TERMINATED",
    "terminatedAt": "2026-07-20T14:30:18Z",
    "coolingOffApplied": false,
    "equipmentDisposition": "PENDING_COLLECTION",
    "equipmentBindings": [
      {
        "equipmentRequirement": "GPON_ONT",

```

```

        "equipmentRef": "ont_01HZ_KAMAU"
      }
    ],
    "settlement": {
      "lineItems": [
        {
          "kind": "PRORATED_REFUND",
          "amount": -1200.00,
          "currency": "KES"
        }
      ],
      "totals": {
        "net": -1200.00,
        "currency": "KES"
      },
      "refundAmount": 1200.00,
      "writeOffMarker": false
    },
    "initiatingActorUserId": "cc-agent-wik-007",
    "initiatingActorRole": "CALL_CENTER"
  }
}

```

Consumers:

- NOT-01 for customer confirmation.
- OSR-RMA-01 EQP for pickup work order when disposition is PENDING_COLLECTION.
- BIL-04 for dunning loop closure when trigger is DUNNING_TERMINATION.
- CVM for churn analytics.
- EM-04 for reporting.

11.2 SubscriptionTerminationRejected

Topic:

```
sophix.subscription.subscription.terminationrejected
```

Emitted after validation, billing, fulfillment, or interruption failure.

12. Drools Package

Package pattern:

```
rules.subscription.terminate.<operator>
```

For WIK:

```
rules.subscription.terminate.wik
```

Required decisions:

Decision	Purpose
validate-preconditions	Reject already terminated subscriptions and invalid state/operation context.
validate-trigger-role	Trigger and actor role are compatible.
resolve-equipment-disposition	Chooses caller-supplied, default, dunning default, or write-off disposition.
compute-cooling-off	Determines whether customer-requested termination is inside cooling-off window.
compute-routing	Determines churn-save and supervisor override routing.

13. Auth

Rules:

- Resolve actor from token, not request body.
- Customer self-service requires operator config enabled.
- Dunning termination requires a BIL-04 service account.
- Contract-expired termination requires scheduler service account.

- Admin termination requires SUBSCRIPTION_ADMIN.

14. Infrastructure Recommendations

Termination does not need separate infrastructure from SUB-WF.

Deploy:

Artifact	Where
Terminate endpoint/controller	SUB-WF API pods.
Termination config migration	Application DB migration pipeline.
Termination BPMN files	Camunda deployment pipeline.
Termination Drools packages	Drools/KIE deployment pipeline.
Termination worker usage	Worker pods/apps from worker DD.

Monitor:

- Termination request count by trigger.
- Termination failure count by rejection code.
- In-flight interruption failures.
- FUL-05 revocation failures.
- HomePass release failures/stub usage.
- Operations stuck in PENDING_TERMINATION.
- Outbox backlog for subscription.terminated.

Medium budget deployment recommendation:

- Run termination API inside existing SUB-WF API pods.
- Run termination workers inside the shared subscription worker deployment.
- Do not create a dedicated termination worker pod unless volume, security, or operational isolation requires it.
- Add alerts for terminal-state failures because manual recovery is harder after network revocation starts.

15. AWS Deployment Recommendations

For AWS, deploy termination as part of the same subscription workflow stack:

Component	AWS recommendation
SUB-WF API	EKS deployment behind internal ALB/API gateway.
Worker apps	EKS deployment with HPA based on Camunda external task lag and CPU.
Camunda	EKS service using the approved foundation Camunda topology.
PostgreSQL	RDS PostgreSQL or Aurora PostgreSQL with Multi-AZ.
Kafka	MSK or approved Kafka platform.
Drools/KIE	EKS service or artifact-driven rule executor.
Secrets	AWS Secrets Manager or Parameter Store with IRSA.
Observability	CloudWatch Container Insights plus application metrics for operation states and worker topics.

AWS alarms:

- PENDING_TERMINATION older than SLA.
- FUL-05 failure spike.
- HomePass release failure spike.
- SubscriptionTerminationRejected spike.
- Outbox backlog for subscription.terminated.

16. Cross-DD Dependencies

DD/module	Relationship
SUB-WF-FRAMEWORK-01	Operation rows, idempotency, interrupt behavior, Camunda start.
DD_SUB-WF-FRAMEWORK-01-WORKERS	Worker topic contracts.
DD_SUB-WF-TERMINATE-01-FLOW-WIK	WIK BPMN implementation.
SUB-LM-01	Subscription state and termination master fields.
BIL-01	Termination intent and settlement calculation.
BIL-04	Calls DUNNING_TERMINATION.
FUL-05	Network/service revocation.
FUL-07	HomePass release.
OSR-RMA-01 EQP	Equipment collection work order after event.
NOT-01	Notifications.
Scheduler	Calls CONTRACT_EXPIRED termination.

17. Build Order

Detailed flow is in DD_SUB-WF-TERMINATE-01-DEV_IMPL_GUIDE-v1.0.md.

Recommended order:

1. Add subscription_termination_config migration and seed rows.
2. Ensure SUB-LM has termination master fields.
3. Add framework operation config for TERMINATE.
4. Implement terminate preview API.
5. Implement terminate trigger API.
6. Implement operation interrupt integration.
7. Implement Drools facts/decisions.
8. Deploy first BPMN, normally WIK.
9. Wire BIL-01 TERMINATION_INTENT.
10. Wire FUL-05 and HomePass release worker.
11. Wire outbox events and notification.
12. Wire OSR-RMA event consumer for equipment pickup.

18. Test Checklist

- [] Termination rejects already TERMINATED subscriptions.
- [] Trigger role gating works for all four triggers.
- [] Customer self-service config is enforced.
- [] Equipment disposition defaulting works.
- [] Dunning equipment disposition/write-off behavior works.
- [] Cooling-off calculation is correct.
- [] Preview does not mutate state.
- [] Termination interrupts in-flight operations.
- [] Success commits subscription to TERMINATED.
- [] Success writes termination master fields.
- [] Success emits SubscriptionTerminated.
- [] Failure emits SubscriptionTerminationRejected.

- [] Notification failure does not roll back successful termination.